

15.1 Ideal Gas Law

Question Paper

Course	CIE A Level Physics
Section	15. Ideal Gases
Topic	15.1 Ideal Gas Law
Difficulty	Hard

Time allowed: 40

Score: /32

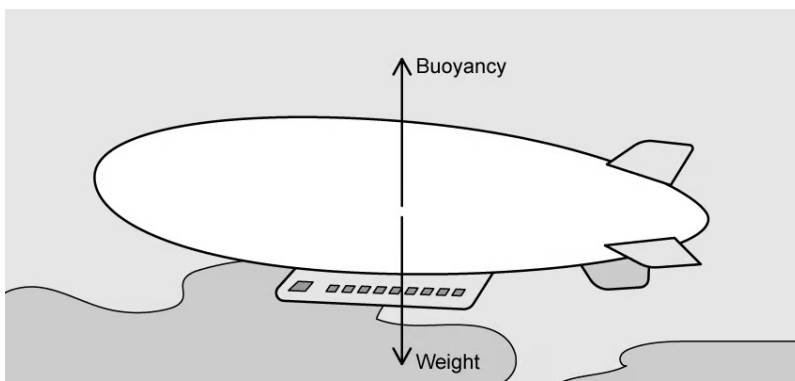
Percentage: /100

Question 1a

An airship floating high up in the Earth's atmosphere is kept in position due to the balance of weight and buoyancy forces.

At one point in the flight, the hydrogen gas has a temperature of 8°C at a pressure of $4.2 \times 10^5 \text{ Pa}$.

The mass of the hydrogen in the ship is 1224 kg.



The atomic mass of hydrogen is $1.00794 \text{ g mol}^{-1}$.

Calculate the density of the hydrogen gas in the airship.

[3 marks]

Question 1b

Calculate the surface area of the inside surface of the airship at this same point in the flight.

[2 marks]

Question 1c

An error with the airship causes it to fall vertically from stationary to the ground. The hydrogen inside has a specific heat capacity of $14.51 \text{ J kg}^{-1} \text{ K}^{-1}$. Due to the nature of the error the Physicists observe that the pressure, volume and specific heat capacity of the hydrogen within the airship remain constant but the temperature changes as it falls.

The mass of the material in the airship is 7320 kg and it hits the ground with a velocity of 200 m s^{-1} .

Calculate the temperature of the hydrogen in the airship at the point just before it hits the ground. Give your answer to as many significant figures as required.

[4 marks]**Question 1d**

(i)

Explain what happened to the airship in part (c)

[5]

(ii)

Explain why there has to be a problem with the data obtained from the airship's fall to Earth.

[1]

[6 marks]

Question 2a

A cylinder of radius 2.5 cm and height 50 cm is fixed with an airtight piston containing an ideal gas of temperature 15 °C. Before the piston is pushed in the pressure is 2×10^4 Pa and the volume is $4 \times 10^{-3} \text{ m}^3$.

The piston is compressed through a height of 20 cm as shown in Fig. 1.1.

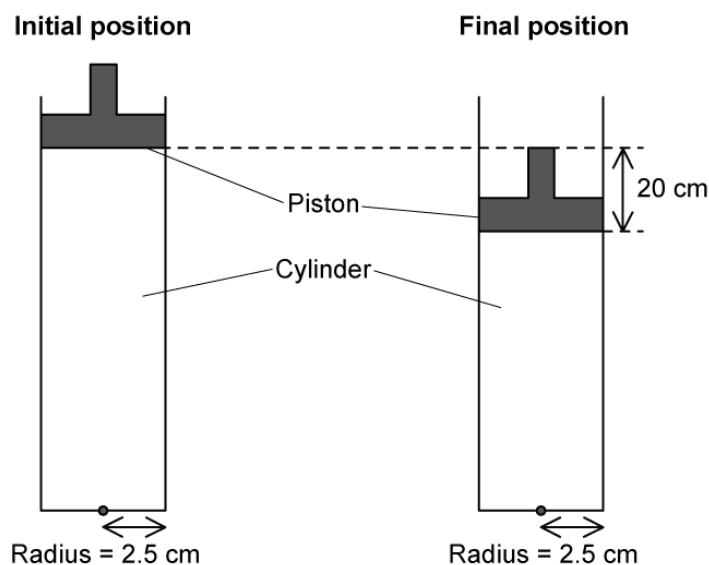


Fig. 1.1

Calculate the change in pressure that takes place in the cylinder.

[3 marks]

Question 2b

It takes 30 seconds to compress the piston in part (a). The cylinder is made of a material such that the surface area on the inside is the same as the surface area on the outside. Assume that each particle exerts an equal force on the sides of the cylinder.

Calculate the change in momentum of the particles upon impact with the sides of the cylinder during the compression.

[4 marks]**Question 2c**

When compressed the particles inside the cylinder have a density of 2.3 kg m^{-3} .

Calculate the root mean square speed (r.m.s.) of the particles inside the cylinder when it is compressed.

[3 marks]

Question 3a

A gas syringe is connected through a delivery tube to a conical flask which is immersed in a beaker of boiling water. The water is being heated by a constant bunsen burner flame as shown in Fig. 1.1. The syringe is frictionless so the gas pressure within the system remains equal to the atmospheric pressure $1.02 \times 10^5 \text{ Pa}$.

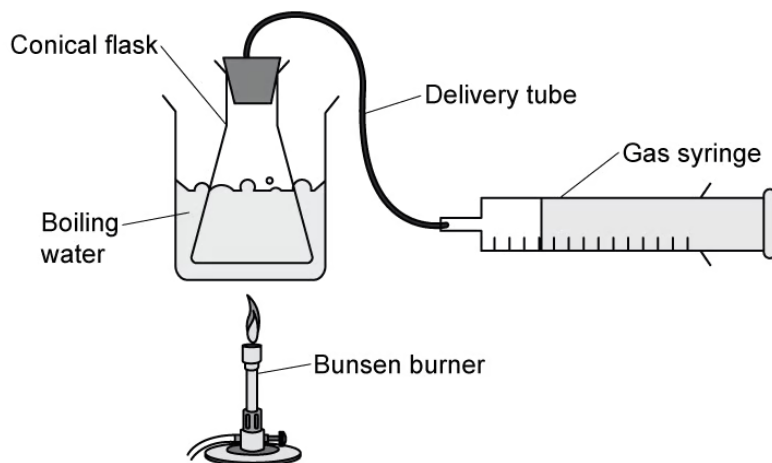


Fig 1.1

The total volume of the conical flask and delivery tube is 325 cm^3 , and after settling in the boiling water the gas syringe has a volume of 30 cm^3 .

Calculate the total number of moles contained within the system.

[2 marks]

Question 3b

The bunsen burner is turned off and the whole system is placed in a freezer. The conical flask is left to cool in the beaker of water which turns into ice.

It takes 5 minutes for all the water to stop boiling and then 3 hours to cool until the point before the water turns to ice. It then takes a further 1.5 hours for all the water to turn into ice around the conical flask.

Sketch, on Fig. 1.2, a graph to show this process.

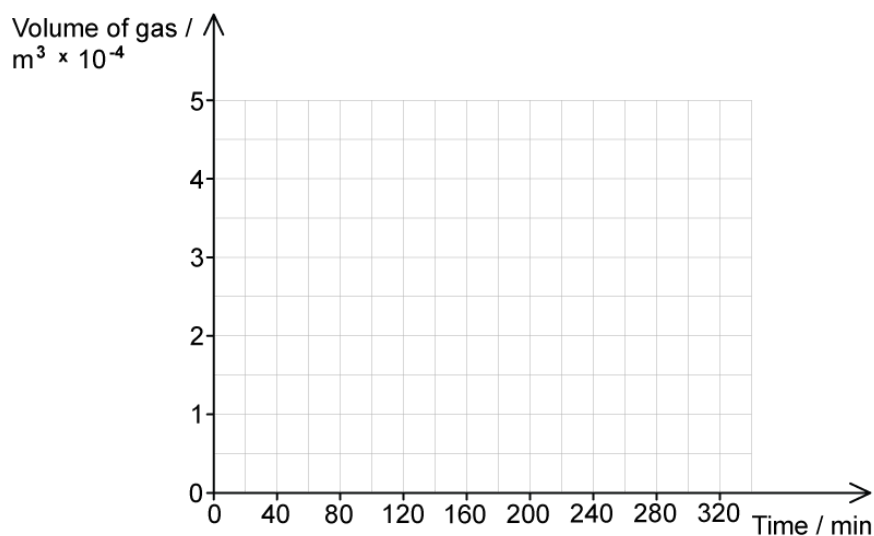


Fig 1.2

[3 marks]

Question 3c

The mass of the ice and the water in the beaker is 240 times the mass of the gas in the system.

Calculate the root mean square speed (r.m.s.) of the particles in the system when the conical flask is surrounded by ice in the beaker in the freezer.

- Specific latent heat of vapourisation = 2160 kJ kg^{-1}
- Specific latent heat of fusion = 324 kJ kg^{-1}
- Specific heat capacity of water = $4084 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

[2 marks]

